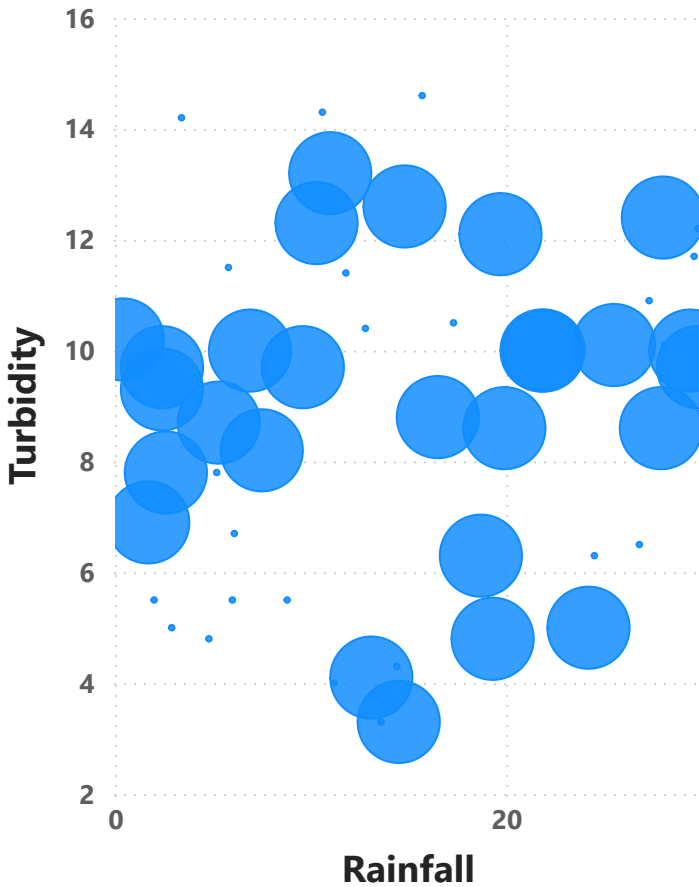
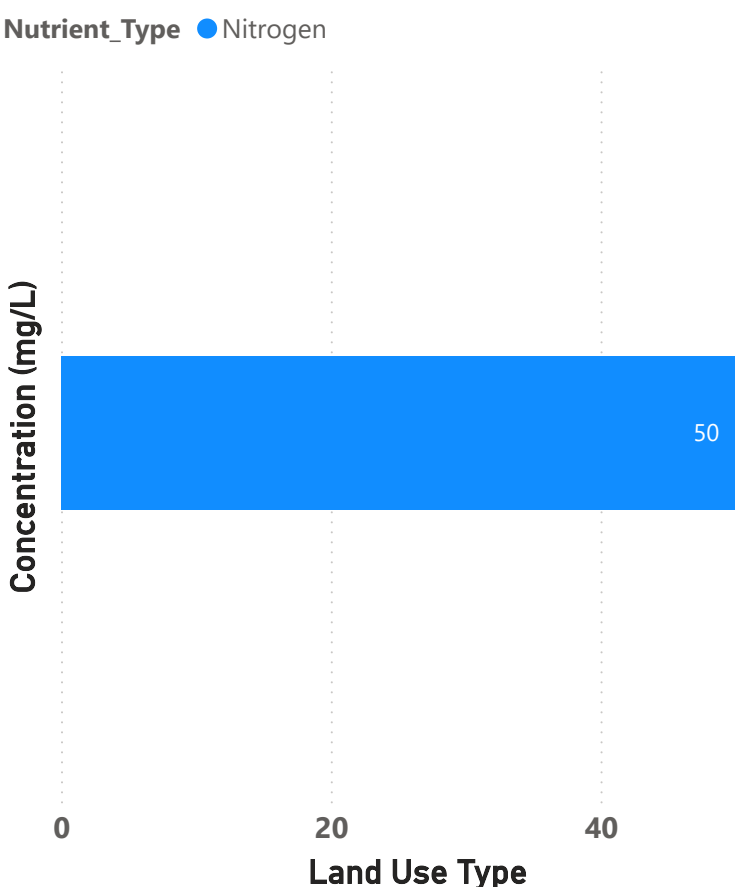


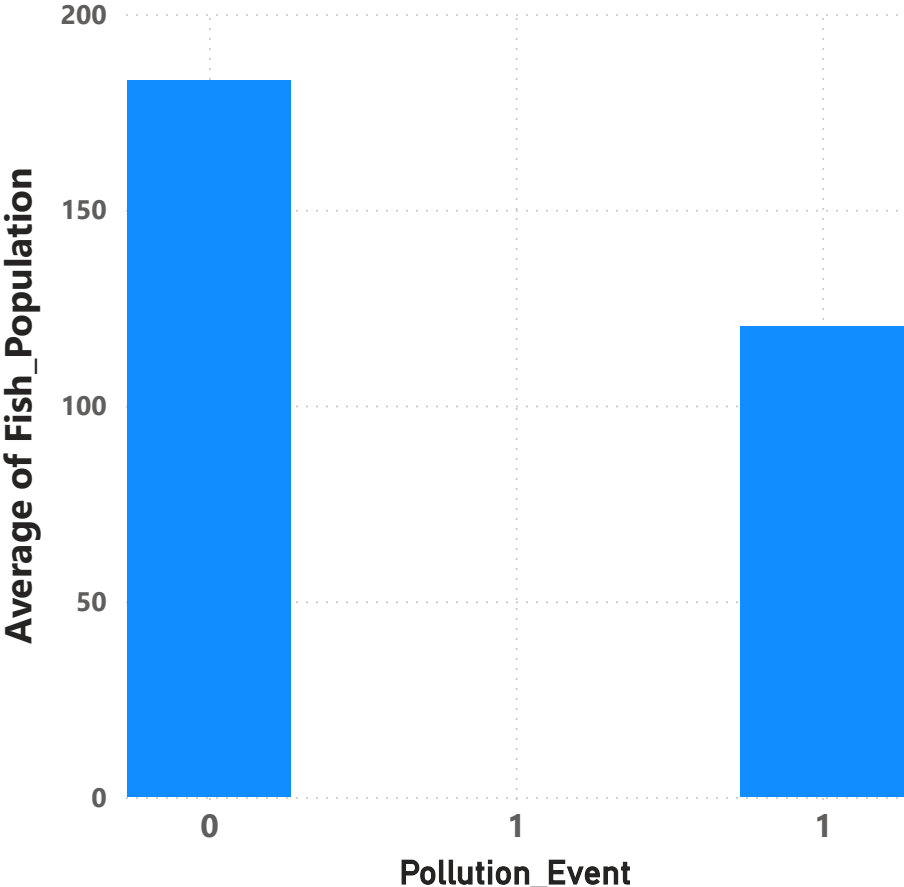
Impact of Rainfall on Turbidity



Impact of Land Use on Nutrients



Impact of Pollution Events on Fish Population



KEY INSIGHTS:

- Rainfall shows a positive relationship with turbidity, indicating runoff-driven sediment transport.
- Agricultural land use is the dominant contributor to nutrient pollution across monitored sites.
- Pollution events significantly reduce fish population, highlighting ecosystem sensitivity to contamination.

IMPLICATIONS:

- Water quality degradation is primarily driven by land-use intensity and rainfall-induced runoff.
- Nutrient loading and sediment transport represent the main environmental stressors in the watershed.

DECISIONS & ACTIONS:

- Prioritize agricultural runoff mitigation through buffer zones and fertilizer management.
- Improve stormwater control systems in urban and agricultural regions.

Date

2025/1/1

2025/2/19

Station_ID

All

Water Quality Metric Selector

☐ pH

☐ Dissolved_Oxygen

☒ Turbidity

☐ Nutrient_Type

☐ Phosphorus

Avg Phosphorus by Land_Use

Land_Use	Avg Phosphorus
Forest	3.7
Agricultural	3.6
Urban	3.1
Other	1.2

3.47

Avg Phosphorus

7.71

Avg DO

8.79

Avg Turbidity

Moderate

Pollution

Pollution Risk

WATER QUALITY INSIGHTS & POLICY RECOMMENDATIONS

EXECUTIVE SUMMARY:

The analysis indicates moderate water quality conditions across the monitored region, with an overall elevated nutrient load reflected by phosphorus levels. Spatial variation shows clear influence from land-use patterns, particularly agricultural and urban activities.

KEY FINDINGS:

- Forest areas consistently demonstrate the lowest phosphorus levels, indicating effective natural filtration and ecosystem stability.
- Agricultural zones exhibit the highest nutrient concentrations, strongly associated with fertilizer runoff.
- Urban areas show moderate but persistent pollution influenced by stormwater discharge.
- Dissolved oxygen levels remain generally within acceptable ecological thresholds, suggesting limited immediate hypoxic risk.

ROOT CAUSE ANALYSIS:

The primary driver of water quality degradation is land-use intensity and anthropogenic activity. Agricultural runoff is the dominant source of phosphorus loading, while urban stormwater systems contribute to diffuse pollution.

Forest ecosystems function as natural buffers, mitigating pollutant dispersion and maintaining baseline water quality stability.

RECOMMENDED ACTIONS:

1. Implement targeted agricultural runoff mitigation strategies, including buffer strips and fertilizer optimization.
2. Upgrade urban stormwater management infrastructure to reduce diffuse pollutant discharge.
3. Prioritize high-frequency monitoring at high-risk stations identified by elevated phosphorus levels.
4. Increase sampling intensity during rainfall events to capture runoff-driven variability.
5. Integrate land-use planning policies to protect watershed-sensitive zones.

DECISION IMPACT:

This dashboard enables evidence-based environmental decision-making by identifying spatial pollution drivers, supporting targeted intervention strategies, and improving long-term watershed sustainability and regulatory planning.